

Mayank Pallai

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github.com/cyprus09 | dev.to/cyprus09

EDUCATION

Nanyang Technological University, Singapore

Aug 2021 – Jun 2025

- B.Eng. in *Electrical and Electronic Engineering*, Specialization: Computer Engineering
- Relevant Coursework: Data Structures and Algorithms, Computer Networks, Computer Architecture, Databases, Software Engineering, Web Application Design, Data Science and AI, AI and Data Mining

WORK EXPERIENCE

Sprinklr

Sep 2025 – Present

Software Engineer

India

- Revamped the contest and events module, live across **40+ countries** for a Fortune 100 client's forums, serving **30M+ users**.
- Sole contributor of the Survey repo, an enterprise form platform (Google Forms-style) serving **15+ enterprise clients**, collaborating directly with global clients and product teams to ship features, processing **60K+ responses/month**.
- Identified inconsistent SDK onboarding and built an LLM-powered pipeline that decomposes requirements, matches our widget library, and auto-scaffolds new widgets for gaps, cutting **onboarding time 40%** and **dev effort 60%**.
- Redesigned gamification engine with admin-configurable scoring rules and optimized **GraphQL** aggregation for multidimensional scoring, scaling to **50K+ users**.

INTERNSHIP EXPERIENCE

Sprinklr

Jun 2025 – Aug 2025

Software Engineer Intern

India

- Engineered internal regression testing tool using **AST**-based parsing via **Babel** and **TypeScript Compiler API**, implementing graph traversal algorithms to generate dependency graphs with **90%+** precision across production-scale codebases.
- Built a high performance system capable of processing **10K+** files, and **60K+** import statements at scale.
- Architected **CI/CD** risk-assessment pipeline integrating **GitLab** workflows with vector embeddings and cosine similarity matching to surface semantic and historical risky changes, reducing post-deployment regressions **20%**.

FileAI

Jun 2024 – Aug 2024

Full-Stack Engineer Intern

Singapore

- Optimised **AWS Lambda** serverless infrastructure reducing costs **15%** through optimised function execution while scaling AI operations with **Bedrock API** for **LLM**-powered backend services.
- Implemented **Python** post-processors delivering tailored AI model outputs for **8** key clients post **OCR** processing.

Switchco Labs

Jan 2024 – May 2024

Software Engineer Intern

Singapore

- Optimized perp DEX trading engine in **Golang**, increasing transaction throughput **12%**, managing **150** open positions across **50+** markets concurrently.
- Identified and migrated read-heavy, frequently accessed data from **PostgreSQL** to **Redis** caching layer, reducing query response time **25%** for high-throughput trading operations.

PROJECTS

LLM Inference Lab [GitHub](#) [Blog](#)

- Built terminal-based LLM inference toolkit (**Qwen2.5/3B**, **PyTorch/MPS**) with per-token entropy tracking and attention-sink detection to characterize context quality and generation confidence.
- Implemented KV-cached speculative decoding and built a unified CLI it with entropy-guided adaptive drafting.

Student Management Portal [GitHub](#)

- Programmed a user-focused web platform as part of Final Year Project, managing events for **800+ MSc** students using **Golang**, **ReactJS**, **Tailwind CSS**, **Supabase**, and **PostgreSQL** and **Docker** for server deployment.

Blockchain Implementation [GitHub](#)

- Engineered **Golang**-based blockchain with **3-node** distributed-system architecture, using the UTXO model for wallet creation, address validation, and secure transactions.

SKILLS

- Programming: C/C++ | Typescript | Javascript | Golang | Python | PostgreSQL | NoSQL | HTML | CSS
- Frameworks & Libraries: ReactJS | Next.js | Tailwind CSS
- Technologies: Git | GitHub | Redis | Docker | AWS | gRPC (Protobufs) | REST APIs | GraphQL | Webpack | Babel |
- Languages: English | Hindi | Odia